

# SS-RESfM: Resume Handbook

State as of September 6, 2026

## 1. Where the project stands

**The paper is essentially complete.** `claude specs/ICLR_ssresfm.tex` is the living draft (ICLR format): all experiments integrated, every table cell machine-verified (`code/verify_iclr_tables.py` — run it, expect 0 mismatches), appendix done, abstract and contributions in findings-first framing, reproducibility + ethics statements in, anonymity clean. Two qualitative figures (`fig_qual_olsson.pdf` = the clean-OOD collapse, the lead visual; `fig_qual_romanforum.pdf` = the causal 60% ceiling) and the contamination-curve figure (`fig_contamcurve.pdf`, trans+rot) are integrated. The experimental program — main 5-seed bands, two-stage, lr grid, hyperparameter sensitivity, and the complete in-distribution contamination curve (0.5/3.1/25/43/60%) — is finished and logged in memory + `claude specs/POLISH_iclr.md`.

**3DV #524 is under review** (frozen; decisions expected ~November 2026). `PATHA_supp.zip` was the submitted supplementary. Post-deadline results are ICLR/ICML-only.

**Companion documents:** `SUMMARY_ssresfm_vs_resfm.pdf` (3-page short summary + the 4-panel figure — the advisor-briefing artifact), `PATHA_3dv_rebuttal_prep.pdf` (rebuttal playbook), `CODEMAP_ssresfm.pdf` (repo changes + per-model code paths), `POLISH_iclr.md` (itemized polish tracker — the authoritative TODO list).

## 2. The submission decision (do this first on resume)

**Chosen plan (Sep 6):** submit to **CVPR 2027** (deadline Nov 7–12; requires withdrawing 3DV first, blind — 3DV notifies Dec 2); read the CVPR reviews at rebuttal stage (~mid-Jan): if rebuttable, stay in (decision late Feb; ICCV ~Mar 5 fallback with full feedback); if clearly fatal (no path to a champion), withdraw and retarget **ICML** (late Jan, private reviews) within the week. The CVPR-formatted paper EXISTS: `CVPR_ssresfm.tex/.pdf` (8pp two-column, fits) + `CVPR_ssresfm_supp.tex/.pdf`; remaining CVPR to-dos: swap in the official CVPR 2027 author kit `cvpr.sty` + real paper ID, port supp tables 15/16-style five-seed material if desired, final read-through. The ICLR tex remains the full-length master — keep edits synchronized. The venue fork below is the superseded analysis, kept for context; it was awaiting Ronen’s read. A drafted email to him (in the chat log; core: 3-page summary + 3 figures, framed via RESfM’s own Table 7) was never confirmed sent — **send it or have that conversation first**. The tree:

- **ICLR 2027** (deadline late Sept 2026): requires withdrawing 3DV first; paper + reviews are *public* during review. *If you resume after late September, this option has expired* and you are automatically on the next branch.
- **Let 3DV decide (~Nov) → rejected:** submit the full strengthened paper to **ICML 2027** (deadline late Jan 2027; reviews private) — this was the recommended risk-dominant route. → *accepted:* publish at 3DV; the post-freeze work (causal in-distribution program, contamination curve, two-stage controls, reliability methodology) is substantial enough to become a *second* paper (“the label

ceiling is causal”) for ICML/ICCV, citing the 3DV paper.

- **ICCV 2027** (~Mar 5) is the fallback after any rejection. CVPR (Nov 7–12) was never reachable while under 3DV/ICLR review.

Key sensitivity: the paper critiques the supervision recipe of the ESfM→RESfM line (Ronen’s own); public-during-review venues expose it to the community — and to the RESfM authors — before acceptance. Privacy was the argument for ICML. Also pending from Fadi: VGPA camera-ready PDF (verify the citation) and the authors’ ~32% 1DSfM tracks — the paper contains a **pre-registered prediction** about those tracks (supervision competitive-or-ahead at ~32%); when they arrive, run both recipes and report the result *whatever it is*. Never soften or remove the prediction sentence.

### 3. Compute that was in flight (will be finished; results NOT yet harvested)

Nothing fires automatically after this session ends. On resume, harvest in this order:

1. **Olsson in-distribution escalation** (2→5 seeds). Six trainings were running: `results/multiscene/{uesfm,resfm}` (20k epochs, babysitter chains attached; a duplicate s22 job was killed — expect exactly one run per dir). When done: fire the standard in-dist eval fleets exactly like s20/s21 — for each seed, SS arms via `run_multiscene_eval.sh` with eval roots `results/crossdataset/uesfm_olssonid_s{22,23,24}_ind_{madweight,` and supervised `resfm_olssonid_s{22,23,24}_ind_eval`, 8 OLTEST test scenes each, queue **waic-risk** (standing policy: ALL jobs). Then update the Olsson column of `tab:indist` (currently 2-seed: SUP  $0.82 \pm 0.37$ ; wttt  $2.31 \pm 0.10$ ; mad  $2.58 \pm 1.14$ ; remove  $2.75 \pm 0.35$ ; weight  $4.46 \pm 0.84$ ), remove the “(Olsson: two seeds)” caveats in the caption/text/figure captions, extend `verify_iclr_tables.py` INDIST expectations, and re-run it. Conclusion at 2 seeds (supervision wins clean in-dist; class-imbalance refuted) is expected to hold — flag prominently if 5 seeds change it.
2. **MegaDepth rebuild program** (Polish list section F). The 667 GB `MegaDepth_SfM_v1.tar.xz` download job (`md_sfm_dl`, script `code/.md_sfm_download.sh`, resumable/idempotent) extracts full per-scene image sets to `code/datasets/raw_megadepth/MegaDepth_SfM/<scene>/`. Verify SFM SCENES: 68 in `.md_sfm_dl.out`. Then: point `code/build_tracks/build_megadepth_rebuilt.py` at the new dir (edit the pool glob from `MegaDepth_v1/.../dense*/imgs` to `MegaDepth_SfM/<scene>`) and submit the 68-scene fleet (v1-dense archive was abandoned: median 43% coverage). Traps already burned: strip `namesList` entries (trailing newline); use the *literal* venv path `u-esfm/.venv38-resfm/bin/python` in `bsub` (never `readlink -f` — it resolves past the venv); clean `.mdrlogs/` between resubmits ( `-o` appends). After tracks build: mirror trainings (`uesfm/resfm` × s20,s21 on rebuilt-27) + evals; **within-rebuild comparisons only** (interpretation ladder in `POLISH_iclr.md` F). Delete `datasets/raw_megadepth` (~0.6 TB) when done.

### 4. Paper polish still open (`POLISH_iclr.md` is authoritative)

- **Page trim — the big one**: main text ends ~p.13 vs. ICLR/ICML’s ~9–10-page limits. Proposed cut order (user never approved — decide): move two-stage control +  $2 \times 2$  diversity-architecture control to appendix (~1.2pp); compress Related Work + classical-baselines/lr-grid pointers (~0.5pp); move `tab:main` rotation/ $N_r$  panels to appendix (~0.8pp); tighten `§whyremoval` (~0.5pp).
- Style swap to the official ICLR 2027 (or ICML 2027) class when targeting; re-measure.
- Abstract option (drafted, not applied): promote the “label-free is the safer default outside the

curated training domain” deployment sentence.

- OpenReview logistics: keywords/TLDR chosen (in POLISH E); arXiv cross-list cs.CV+cs.LG.
- Decide fate of exploratory figures not in the paper: `fig_indist_vs_ood.pdf`, `fig_4panel_{faith,scratch,released}` (candidates for appendix; the 4-panel scratch version is embedded in the Short Summary).
- Final passes: rerun `verify_iclr_tables.py`; grep compiled PDF for ?? refs; anonymity re-check after any edits.

## 5. Operational facts you will otherwise re-derive painfully

- **Queue policy:** ALL jobs (trainings and evals)  $\rightarrow$  waic-risk. Eval bsub pattern: `-gpu "num=1:j_exclusive=yes:gn -R "rusage[mem=50000]"`; trainings via `multiple_scenes_learning.py --conf ... --phase TRAINING` with babysitter `trigger_ablation_resume.sh` (see `resume_stalled_trainings.sh` lines 110–135 for the canonical block). `bsub` under `mbatchd` throttling retries internally — *check for duplicate submissions* (it double-submitted once).
- **Root naming:** main-band SS arms `uesfm_finelr[_s2X]_rf_{arm}_{ds}_eval` (MegaDepth: `uesfm_finelr[_s2X]_baseline` `resfm_finelr[_s2X]_...`; released ckpt = `resfm_repro_env38` (MegaDepth; 0.209) and `resfm_repro_{ds}` (OOD) — `resfm_repro` bare is a stale variant (0.286), do not use; `faithful = resfm_faithful[_s2X]_...`; `in-dist = {uesfm,resfm}_{pool}_s{s}_ind[_arm]_eval`, `pools olssonid/bmvaid/ldonly/hardonly/ldsfaid` (`_indh_` = merged-pool hard test).
- **Reading results:** per-scene file `<root>/<scene>_ba/Results_FINE_TUNE_stage_1_esfm_outliers_shallow.xlsx`, post-BA translation `ts_ba_final_mean/_med`, rotation `Rs_ba_final_mean`, cameras `#registered_cams_final`. Bands = `mean $\pm$ std (ddof=1)` over training seeds of per-root scene-means. The recorded translation metric is *not* reproducible as naive camera-center distance from the `forFigures` npz (alignment/normalization differs); for camera-map figures extract positions from the eval’s own `plots/Final_plots_final` (plotly traces `cam_gt/cam_learn`) — that is how `fig_qual_olsson.pdf` was made. `registered_cams_indices` in the npz is sometimes a 0/1 *mask*, sometimes an index list — check before pairing.
- **Compiling:** `tectonic` at `~/.local/share/mamba/envs/tex/bin/tectonic -X compile <file>.tex` from `claude specs/`. Figures regenerate from `scratchpad` scripts; the aggregated numbers behind the 4-panel figure are in the scripts’ JSON (regenerate by re-running the aggregation against the roots above if lost).
- **Standing hygiene:** after each model’s eval fleets complete, prune its `models_all` to `best+final+reselection` winners (`prune_checkpoints_dryrun.py`); lean eval outputs are default (diagnostics only via `eval.save_eval_diagnostics = True` appended as an `eval {}` block to a generated conf, with a `_diag_eval` root).
- **Memory:** the assistant’s memory index (auto-loaded each session) holds the verdict log — `fiveseed-bands-and-30-indist-contamination-curve`, `megadepth-rebuild-blocker`, `paper-framing-doctrine` (never sell the delta; the findings are the paper), `phase0-spec-regime` (naming/metric rules). Trust it as the starting point; verify anything file-specific still exists.

## 6. Thirty-second resume checklist

1. Which venue window is open? (§2) — talk to Ronen if not yet done; 3DV decision known?
2. `bjobs`; harvest Olsson escalation  $\rightarrow$  fire evals  $\rightarrow$  5-seed `tab:indist` column (§3.1).

3. Check MegaDepth-SfM extraction  $\rightarrow$  rebuild fleet  $\rightarrow$  mirror trainings (§3.2).
4. Fadi's tracks arrived?  $\rightarrow$  run both recipes, report the pre-registered result.
5. Page trim + venue style swap + final verify run (§4)  $\rightarrow$  submit.